Michael Murray

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MANUFACTURING PROCESSES

As a mechanical design engineer, I treat design for manufacturing as paramount: a design is only as good as the process that can produce it. I spent roughly 3 years as a manufacturing engineer at a premier plant for the world's leading metal-cutting tool company, and I machined parts and build my own prototypes in my former employer's shop. The hands-on habit started early — I grew up in the garage as the son of an engineer, and I owned and operated an LLC handyman business through high school and part of university. Whenever schedules and regulations allow, I personally build, test, and troubleshoot my prototypes end-to-end, including the labor, planning, and documentation (and, admittedly, the frequent troubleshooting). I always have at least 2 personal projects going at a time.

The list below was compiled from memory and omits the most basic processes (hand drills, hammers, and the like). It covers processes I have personally performed, designed for, or witnessed in production.

Designed for and physically performed	Designed for and witnessed in production	Finishing and post-processing
Machining • CNC machining — programming (CAM) and running: milling, turning, drilling, tapping, grinding • Manual mill and manual lathe • Drilling, reaming, boring • Surface grinder, bench grinder, angle grinder • Most any form of cutting of most materials **Fabrication and assembly** • Welding (mostly arc and stick) • Press brake, hydraulic and arbor press • Deburring • Wiring and controls (Arduino, Raspberry Pi, PLC, myriad controllers) • Any and every hand tool **Additive** • 3D printing — FDM, SLA, and SLS; printer maintenance and modification • Materials: PLA, ABS, TPU, TPE, PETG, PA-CF, PC, resins, PA12 and PA6 powders • Silicone molding of prototype parts **Inspection** • CMM and other dimensional inspection	**Metal forming and joining** • Sheet metal — bending, stamping, deep drawing, roll-forming • Extrusions • Thread forming • Weldments (fillet weld sizing per AWS D1.1) • Brazing • Piping **Casting and molding** • Die casting and investment casting • Injection molding and overmolding • Thermoforming **Material removal** • Wire EDM • Laser cutting **Additive (production)** • DMLS, DLS, SLS, MJF / PJF **Other** • Heat treatment • Insulation	**Coatings and plating** • Powder coating • Painting • Anodizing • Zinc-flake coating (Geomet, ISO 10683) • Hot-dip galvanizing • Epoxy coating **Surface preparation** • Sand / media blasting • Sanding • Acetone bath (vapor smoothing) **Other** • Autoclaving